

Speech that Protects a Model

Belief updating, public concession, and access to evidence — three mechanisms to distinguish

Reality tells the system: “Your model is wrong.” The system answers: “So much the worse for reality.”

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github.com/muqali2816/
mindloom_IWAI_2026_speech_protects_model

1 The question

Active-inference accounts explain how dialogue *aligns* generative models (Friston & Frith 2015; Friston et al. 2020). We ask the converse: how can an exchange continue for many turns and leave a significant expectation **unrevised** — and what would speech have to make observable for that protection to be *measured* rather than presumed?

2 One situation, two trajectories

Constructed example; not an observed dialogue.

Shared opening. Alex: “I sent the revised budget with the latest prices.” Sam: “The attachment is version 3, with the April prices; version 4 has the May prices.”

A — verification invited

Alex: “Can we compare the files? I thought I had attached version 4.”
Sam shows the attachment.
Alex: “You’re right; I sent the older file. I’ll check next time.”

B — verification restricted

Alex: “You always look for mistakes in my work.”
Alex: “I know what I sent. If you trusted me, you would stop questioning this.”
Sam: “There is no point showing you the file.”

Two coordinates differ at once: *what is said* (a concession is or is not made) and the *form* of the moves (A invites joint verification, B restricts it). B does **not** reveal whether Alex privately changed their mind. Three candidate mechanisms to distinguish:

- ▶ **reduced uptake** ($\omega < 1$): received evidence has a weaker effect on private belief
- ▶ **costly expression** ($c > 0$): private belief can change while public concession remains costly
- ▶ **restricted access** ($\gamma > 0$): updating from received evidence is intact, but fewer checks occur

These mechanisms cannot be identified from this exchange alone.

3 Part I — what speech makes observable

Mindloom annotates a speaker’s contribution in its supplied context, with an explicit evidence trail or abstention when evidence is insufficient (L1 evidence atoms → L2 features → L3 evidence requirements → L4 configuration → L5 typed output). Eight descriptive configurations, one closure function (SEAL) and one transition event (SHIFT); these labels do not establish intent or a hidden psychological mechanism:

BUILD	joint elaboration or repair, response still available
SEEK	inquiry that makes an answer relevant
UNSEAL	first-person disclosure, no regulation demanded
LOCK	language constraining the addressee’s replies
DRAIN	other positioned as required source of support
FLOOD	intensification or repetition dominating the exchange
EDGE	alternation between approach and withdrawal
VOID	withdrawal from an offered opportunity to respond
SEAL (function)	closure of a topic or channel
SHIFT (event)	change with before-and-after configuration evidence

Closure (SEAL) alone does not establish defensiveness. In the simulation below, SEEK and LOCK serve as proxies for invitation and restriction of verification; this mapping remains to be tested in real dialogues.

Benchmark: 589 English monologue fragments generated for the ontology and fixed before any run; Tier A 169 cases (35–71 words, no near-duplicates); Tier B 420 (15 near-duplicate pairs). Provenance and annotator separation incompletely documented. Internal evaluation on ontology-generated fragments; performance on naturally occurring dialogues remains untested.

Re-scored July 2026 archive (169 cases, majority of three stored runs, no new model calls):

system	correct	Macro-F1	SHIFT
direct prompt (Claude Haiku 4.5)	145/169	0.837	18/20
typed engine	129/169	0.757	2/20
keyword rules (tuned on these cases)	43/169	0.195	0/20
majority class (BUILD)	20/169	0.021	0/20

In this archived comparison, the direct prompt outperforms the typed engine overall; the engine’s proposed value is its inspectable evidence trail. Six raters assessed features, not the ten labels; reported Fleiss’ κ : 1.00 for lexically direct features, 0.23, 0.13 and −0.11 for selected non-lexical features; rater matrices are unavailable. A gate ablation at fixed model and prompt was *not* run.

Reused below: the engine’s confusion rows for gold LOCK (10 LOCK, 3 SEAL, 2 BUILD; 15 cases) and gold SEEK (20/20) define an assumed noise channel for *restrict* / *invite*; the transfer of these small-sample rows to real dialogue forms is unvalidated.

4 Simulation-based group comparisons: dialogues per group for ≈80 % power

Welch test, $\alpha = 0.05$. Smallest tested group sizes reaching approximately 80 % power under the simulated effects; not sample-size recommendations for human studies. > 400: target power not reached at the largest tested group size.

	atten. vs cost	atten. vs access	cost vs access
non-concession rate, rounds 9–12	200	35	25
absolute private-belief change	17	25	> 400
restriction rate	> 400	12	12
listener’s model-based $P(\omega_A=0.2)$	12	12	> 400
listener’s model-based $P(c_A=1.6)$	25	50	12

Different measurements distinguish different mechanism pairs efficiently; no single listed measurement reaches the target power for all three pairs within the tested range.

5 Part II — public non-concession alone does not identify the mechanism

One speaker, binary fact x , prior $q_0 = 0.75$ on its own position; evidence e of reliability $\rho = 0.75$ arrives with rate $\lambda_0 = 0.35$ (0.90 after ASK). Subjective likelihood tempered: $A_\omega(e \mid x) \propto A(e \mid x)^\omega$, so

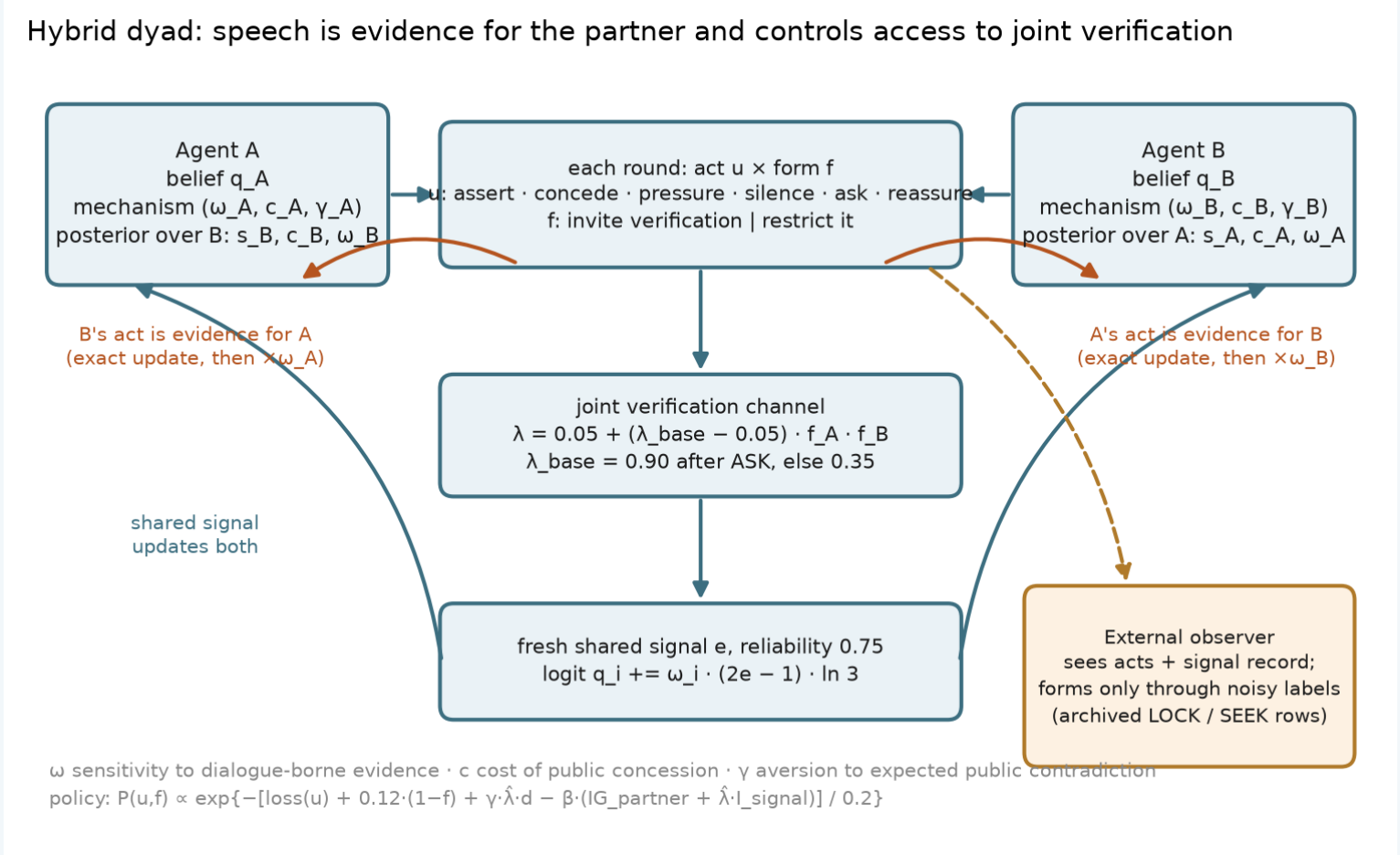
$$\text{logit } q_t = \text{logit } q_0 + \omega m_t \ln \frac{\rho}{1-\rho}, \quad G(u) = \ell(u; q_t, c) - l_\omega(q_t, u),$$
$$P(u) \propto e^{-G(u)/\tau}, \quad \ell(\text{ASSERT}) = k(1 - q_t),$$
$$\ell(\text{CONCEDE}) = kq_t + c, \quad \ell = b \text{ for PRESSURE, SILENCE, ASK; } k = 2, b = 1.10, \tau = 0.20.$$

Matched on late non-concession (ω, c) = (0.10, 0) vs (1, 1.4); 4 000 shared worlds:

	reference	attenuation	cost
non-concession, rounds 9–12	0.654	0.989	0.990
final prob. of true state	0.797	0.538	0.809
absolute belief change	0.349	0.051	0.356
late ASK rate	0.041	0.050	0.150

Nearly identical late non-concession rates, but different private belief updating. Removing c from round 7 raises the fraction of cost-agent dialogues with at least one late concession from 0.061 to 0.433; the attenuation condition remains at 0.054. Removing information gain while retaining all five actions reduces the between-mechanism ASK-rate gap from 0.099 to 0.083; the estimated contribution is 0.017, about 17 % of the full gap. Part II results are taken from the previously checked simulation archive; they are not human measurements.

6 Bridge — a dyad in which speech is evidence and controls access



Both agents infer; each move is an **act** $u \in \{\text{ASSERT, CONCEDE, PRESSURE, SILENCE, ASK, REASSURE}\} \times$ a **form** f ($f = 1$: invite; $f = 0$: restrict). Assumed verification rule: both agents must invite verification for the shared-evidence channel to remain open above its floor:

$$\lambda_t = 0.05 + (\lambda_{\text{base}} - 0.05) f_A f_B, \quad \lambda_{\text{base}} = 0.35 \text{ (0.90 after ASK),}$$

so one restricting form reduces the arrival rate to the floor 0.05, not to zero. Policy over act $\times$ form:

$$G(u, f) = \ell(u) + \delta(1 - f) + \gamma \lambda(u, f) d_t$$
$$- \beta[\lambda(u, f) l_\omega + \text{IG}_{\text{partner}}(u)],$$
$$P(u, f) \propto e^{-G(u, f)/\tau},$$

$\delta = 0.12$, $\beta = 1.5$, $\tau = 0.2$; γ weights the expected cost of a shared signal contradicting the speaker’s initial position (d_t); c prices each public concession. The partner’s act is evidence through a level-0 model of the partner. Each update starts with an exact Bayesian posterior; ω then scales the change in its x -marginal — this differs from the likelihood tempering of Part II. The signal term l_ω is a stated heuristic (error ≤ 0.014 nat at $\omega = 0.2$).

Model-based external observer: sees acts and shared signals; receives forms only through noisy labels drawn from the LOCK / SEEK rows; computes the exact posterior over the speaker’s mechanism family. Null channel: labels drawn from the *act* instead of the form. Oracle: the simulation’s true forms. Ten implementation checks pass (incl. act-for-act reproduction of the M-bridge prototype and the reciprocal dyad to $3 \cdot 10^{-12}$); these establish numerical consistency, not validation in human dialogues.

7 What noisy form labels add to model-based inference

Share of simulated dialogues in which the observer’s most probable mechanism family is the generating family (argmax of the posterior after round 12; 2 000 dialogues per generator):

generator	acts+signals	+noisy labels	+true forms	+null labels
attenuated	0.612	0.672	0.678	0.612
costly	0.690	0.765	0.785	0.690
access-protecting	0.681	0.756	0.812	0.681

- ▶ **Mechanism family:** noisy form labels ($\eta = 0.25$) raise this share by 0.060–0.075 (table); the mean posterior on the generating family rises by 0.013–0.141, most for access cost. Neither is a classification accuracy for people.
- ▶ **Next signal:** noisy labels provide 53–61 % of the log-loss gain obtained from true forms, under the specified noise model ($\eta = 0.25$).
- ▶ **Next act:** labels add at most 0.006 nat. Null labels generated from already observed acts add no information under the specified model.

This gain depends on the stipulated form-to-access rule and the assumed label-noise channel; both require validation in real dialogues.

8 Three mechanisms compared across measurements — speaker A with a reference partner

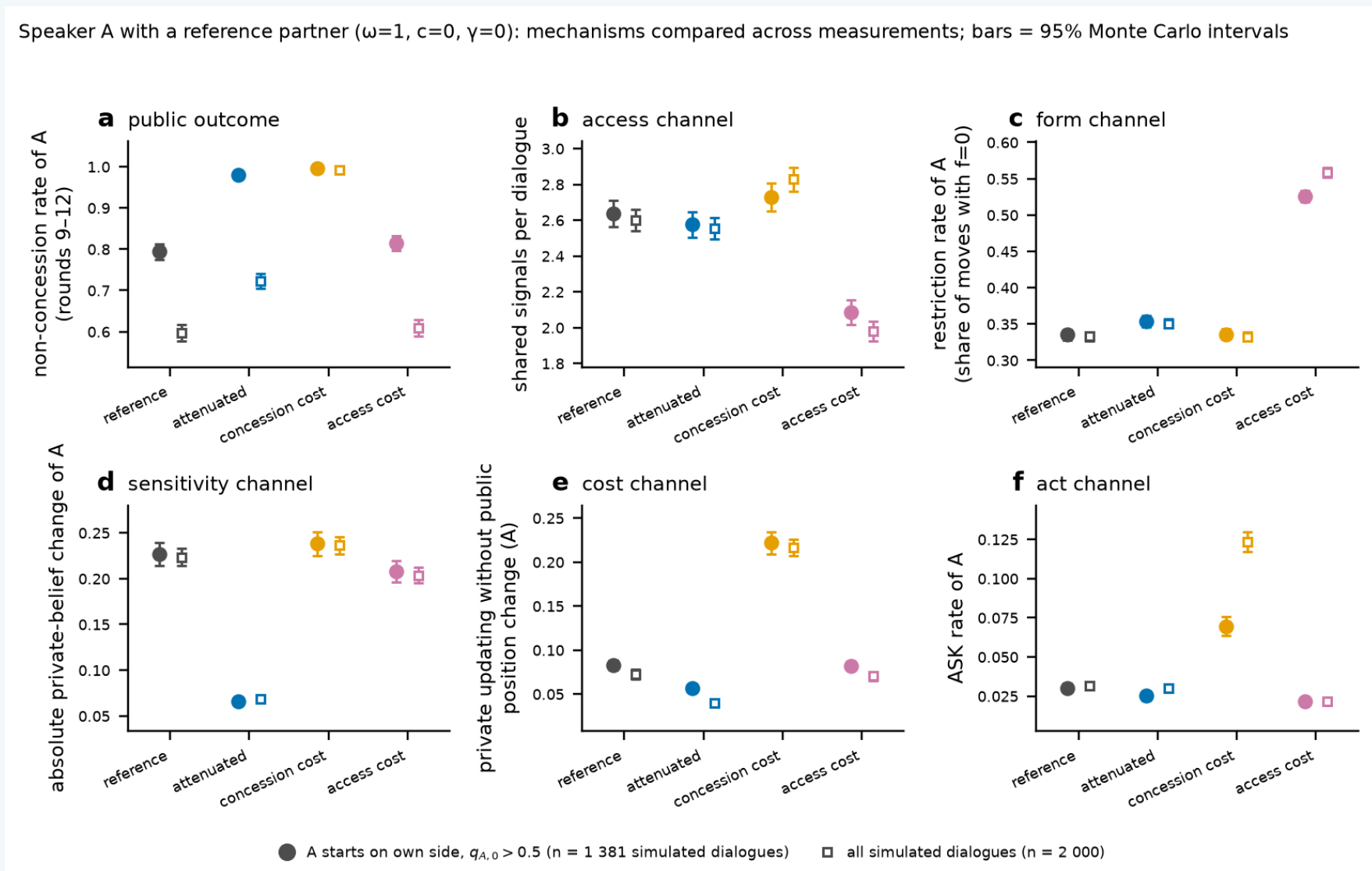


Table: subsample in which A starts on its own side ($q_{A,0} > 0.5$; $n = 1381$ simulated dialogues); the figure also shows all 2000. “Private updating, public position unchanged” = $|q_{A,12} - q_{A,0}|$ if A’s public position at round 12 equals its initial one, else 0; mean over dialogues.

measurement	reference	$\omega=0.2$	$c=1.6$	$\gamma=2$
non-concession rate, rounds 9–12	0.79	0.98	1.00	0.81
absolute private-belief change	0.23	0.07	0.24	0.21
private updating, public position unchanged	0.08	0.06	0.22	0.08
shared signals per dialogue	2.64	2.58	2.73	2.09
restriction rate	0.34	0.35	0.34	0.53

Attenuation and concession cost produce similarly high non-concession rates, but different private belief changes: evidence has less influence under attenuation, while costly concession suppresses public acknowledgment despite private updating. The access-cost condition has a non-concession rate close to the reference condition, but more restriction and fewer shared signals; updating from received evidence remains intact. This formalises a possible access-restriction mechanism illustrated by trajectory B.

9 Responses to interventions in the specified simulation

- ▶ For the concession-cost condition, a partner REASSURE at round 8 increases the fraction of dialogues with at least one concession in rounds 8–12 by 15.5 percentage points relative to placebo (a REASSURE whose cost relief is disabled); removing c from round 8 increases it by 27.5 points. Both contrasts are exactly zero in the other conditions, where $c = 0$ by construction.
- ▶ In the access-cost condition, removing γ from round 8 increases shared signals in rounds 8–12 by 0.199 per dialogue and reduces the restriction rate in rounds 8–12 by 18.7 percentage points; exactly zero where $\gamma = 0$ by construction.
- ▶ In this implementation, one strong shared observation (reliability 0.95, round 8) increases the probability assigned to the true state least under attenuation: +0.058 versus +0.09–0.12. This ordering changes with where attenuation is applied.
- ▶ Setting the partner’s form to invite from round 8 increases shared signals least in the access-cost condition (+0.13 versus +0.31–0.33 per dialogue); the focal agent increases restriction (+0.067).

Listener’s model-based estimates — not an intervention. Partner B’s posterior at round 12 over A’s parameters (uniform prior over $c \in \{0, 0.8, 1.6\}$ and $\omega \in \{0.2, 0.6, 1\}$, i.e. 1/3 each): $P(c_A=1.6) = 0.45$ in the concession-cost condition (0.23–0.32 elsewhere); $P(\omega_A=0.2) = 0.59$ in the attenuation condition (0.42–0.47 elsewhere). Means over the same $n = 1381$ dialogues.

10 Scope and limits of the evidence

- ▶ All bridge results come from simulations of specified mechanisms; no mechanism was fitted to or validated against human dialogue data.
- ▶ The form-to-access rule and the transfer of annotation confusion rows to dyadic forms are modelling assumptions requiring empirical validation.
- ▶ An independent implementation retains the qualitative access effect. Attenuation effects depend on where evidence weighting is reduced; with attenuation applied only to shared signals, the agent resembles the reference condition.
- ▶ In the archived comparison, the direct prompt outperforms the typed engine overall. Keyword rules were tuned on the scored cases; no gate ablation at fixed model and prompt was run.
- ▶ In the tested reciprocal simulation, adding a regime variable provides essentially no improvement in next-act prediction over a last-act baseline. Physiological or allostatic costs were not modelled or measured.

11 Next empirical test in real dyads

Measure private belief before and after dialogue using confidential reports; record public positions, invitations and restrictions of verification, actual checks and evidence exposure, and the partner’s estimate of the speaker. Private-belief reports are measurements, not direct access to belief. Test whether invite/restrict annotations predict subsequent verification and belief change beyond action counts and prior context, on held-out dyads. Use dialogues sampled independently of the ontology and a frozen coding manual; report inter-rater reliability. Annotators should be blind to subsequent verification and belief reports. The proposed annotation-to-access link would be weakened if annotations add no predictive value beyond actions and context on held-out dyads. Randomise acknowledgment costs to test intervention responses; determine human sample sizes from pilot data.